AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 25 · TEACHER GUIDE

Sequence representations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make every numeric value carry its schema.

Separate token identifiers from measurements and masks;keep literal markers,unknown symbols and missing data distinct.

Keep one habit: Assess independent IDs,mask,last real index and correct sensor denominators using checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Encode an ordered **sequence** with an explicit vocabulary

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Pooling discards detail. A maximum can preserve the largest value while losing its location and the other values; identical summaries do not imply identical **input**s.

Opening retrieval

Without opening the old lesson, explain one key idea from Augmentation and leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Check whether a transformation preserves the task label

Reference example, not the only acceptable wording: Teacher explanation: Original d = -2 and class 0. Row swapping leaves d = -2 and class 0. Column swapping gives rows [1, 0], [1, 0], d = 2 and class 1.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Look for: Compute max and mean pooling under an explicit window rule

Reference example, not the only acceptable wording: Teacher explanation: Values are 1, 3, 2 and 7. Sum is 13, maximum 7 and mean 3.25. Each pooled **output** has shape 2 by 2.

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an **input** through a hidden representation and a decision rule

Reference example, not the only acceptable wording: Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. **Prediction**s are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

Pictured end goal

By the end, I can explain **sequence** representations using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Augmentation and leakage

Pooling and spatial information

Learned representations

For the compact alternative schedule, use two 45-minute sessions. Arrange event cards physically before using IDs. Keep **token id**entifiers separate from the sensor measurements used for meaningful mean calculations.

The encoder is a fixed classroom schema, not a learned language model. All indexing is zero-based and all **mask**s in the examples are contiguous right-**padding mask**s.

Outcomes

Encode an ordered **sequence** with an explicit vocabulary

Use **padding** and a **mask** without inventing real events

Identify information lost by counts unknown symbols and truncation

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach token names,real positions and measurement values distinctly

> Keep main GO/TURN/STOP,pictured LEFT/RIGHTand independent OPEN/CHECK/CLOSEschemas separate.

Main definitions,complete encodings and guided key

Use fictional robot event logs whose symbols are GO, TURN and STOP. The task asks whether the final real event is STOP. This is a representation exercise, not a claim that an event log proves a robot is safe.

GO then STOP and STOP then GO contain the same symbols but have different final events.

A count vector discards order, so no deterministic rule using only the first two count vectors can distinguish their answers. A **sequence** of IDs preserves order if we retain each position.

Declare the vocabulary: PAD = 0, UNK = 1, GO = 2, TURN = 3 and STOP = 4. PAD is reserved for added storage positions; UNK represents a real but unrecognised symbol.

IDs are names, not measured amounts. STOP having ID 4 does not make it twice as much as GO with ID 2. Changing the vocabulary requires changing every consumer consistently.

Do not interpret numerical distance between IDs as semantic similarity.

Store each log in four positions, **padding** on the right. A **mask** value 1 marks a real position and 0 marks **padding**. The encoder rejects empty logs, logs longer than four, and literal reserved markers.

Other unrecognised symbols become UNK and still receive **mask** 1.

The code assumes a list of symbol strings and uses exact spelling. It does not fit a vocabulary from data.

If a future project fits vocabulary choices or frequency cutoffs, derive those choices from **training** data and apply the saved encoding consistently to validation and final cases.

Complete input log	Four stored IDs	Real-position mask	Final real position / meaning
GO STOP	[2, 4, 0, 0]	[1, 1, 0, 0]	1 /STOP
STOP GO	[4, 2, 0, 0]	[1, 1, 0, 0]	1 /GO

Complete input log	Four stored IDs	Real-position mask	Final real position / meaning
TURN STOP	[3, 4, 0, 0]	[1, 1, 0, 0]	1 /STOP
WAIT STOP	[1, 4, 0, 0]	[1, 1, 0, 0]	1 /STOP; WAITidentity lost at real position 0
GO TURN GO STOP	[2, 3, 2, 4]	[1, 1, 1, 1]	3 /STOP; no padding

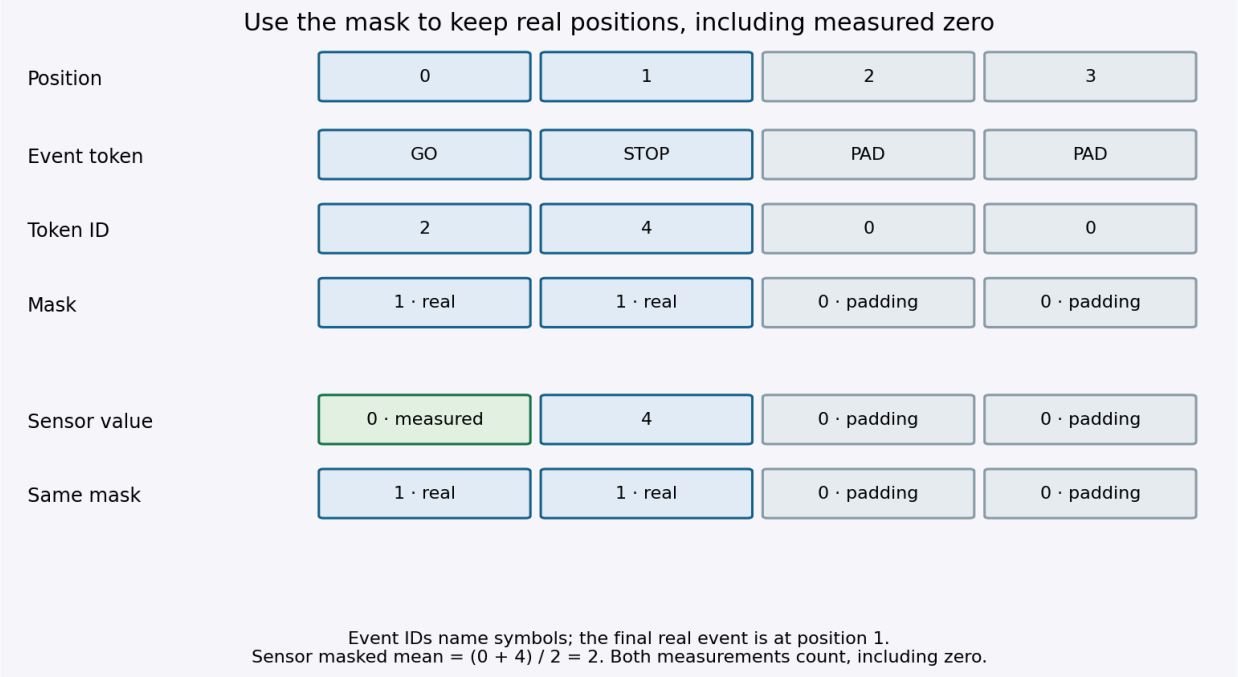
The encoder uses PAD0, UNK1, GO2, TURN3, STOP4 and exactly four stored positions with right padding. Unknown actual symbol strings consume a real position and get mask 1. Missing input is a different problem:the function assumes a list of actual strings and does not validate every type. Literal reserved marker strings PAD/UNK, empty logs and more than four real events are rejected.

The appended visible examples print IDs and masks; they do not train a network or infer event safety.

Guided: TURN STOP becomes IDs [3, 4, 0, 0] and **mask** [1, 1, 0, 0]. Length is 2, so final real position is 1 with ID 4, STOP. Position 3 is PAD.

Exit: the full five-event log answers Yes, but its first four events end in TURN and answer No. Rejection makes the unsupported length visible and avoids silently changing the question.

Aligned masks and every sensor denominator



Complete sensor calculation	Selected values / denominator	Result
Stored[2, 4, 0, 0], mask[1, 1, 0, 0]	Real[2, 4], divide by2	3
Wrong all-four-slot average	(2+4+0+0)/4	1.5
Stored[0, 4, 0, 0], same mask	Real[0, 4], divide by2	2
Wrong discard-every-zero policy	Only[4], divide by1	4
Wrong all-four-slot average for[0, 4, 0, 0]	4/4	1
All-zero mask	No real measurements	Reject or use an explicitly declared empty policy

Separate pictured vocabulary and real-zero scope

Distinguish **padding** positions from meaningful zero sensor readings.

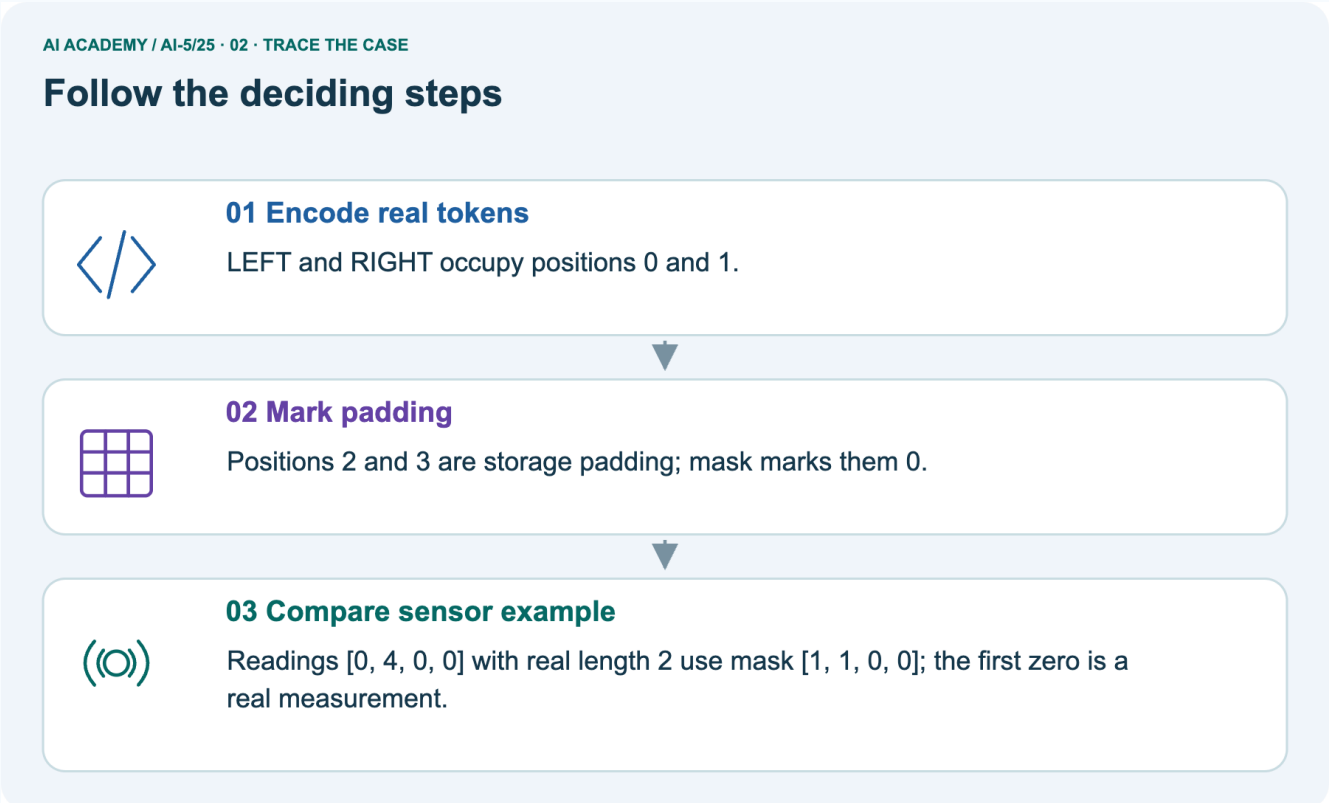
Toy tokens PAD 0, UNK 1, LEFT 2, RIGHT 3; stored length 4. **Sequence** LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture uses a separate vocabulary:PAD0, UNK1, LEFT2, RIGHT3 and four stored positions. LEFT RIGHTgives IDs[2, 3, 0, 0], mask[1, 1, 0, 0]. Its sensor example[0, 4, 0, 0]uses the same mask because the first two measurements are real. The first zero is measured; the last two are storage placeholders. Removing every numeric zero would wrongly discard that measurement.

Keep this LEFT/RIGHTvocabulary separate from the main GO/TURN/STOPvocabulary; the same IDnumber has meaning only under its declared mapping.

Trace the value-only padding error



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Encode real tokens	LEFT and RIGHT occupy positions 0 and 1.
2. Mark padding	Positions 2 and 3 are storage padding; mask marks them 0.
3. Compare sensor example	Readings [0, 4, 0, 0] with real length 2 use mask [1, 1, 0, 0]; the first zero is a real measurement.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

Change: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example.

Prediction: The first real sensor reading is incorrectly discarded.

Reason: Its numeric value is 0, but its position is real according to the declared length.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Sequence	An ordered list of elements in which positions can matter
Token ID	A numerical identifier assigned to a symbol by a vocabulary
Padding	Placeholder positions added to make stored sequences the same length
Mask	A list marking which stored positions are real inputs and which are placeholders

A padded log still has two real events

STORED POSITION	TOKEN ID	MASK
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A padded log still has two real events

STORED POSITION	TOKEN ID	MASK
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

Encode TURN STOP, give its mask and real length, then identify the final real position and its ID. Why would reading stored position 3 give the wrong event?

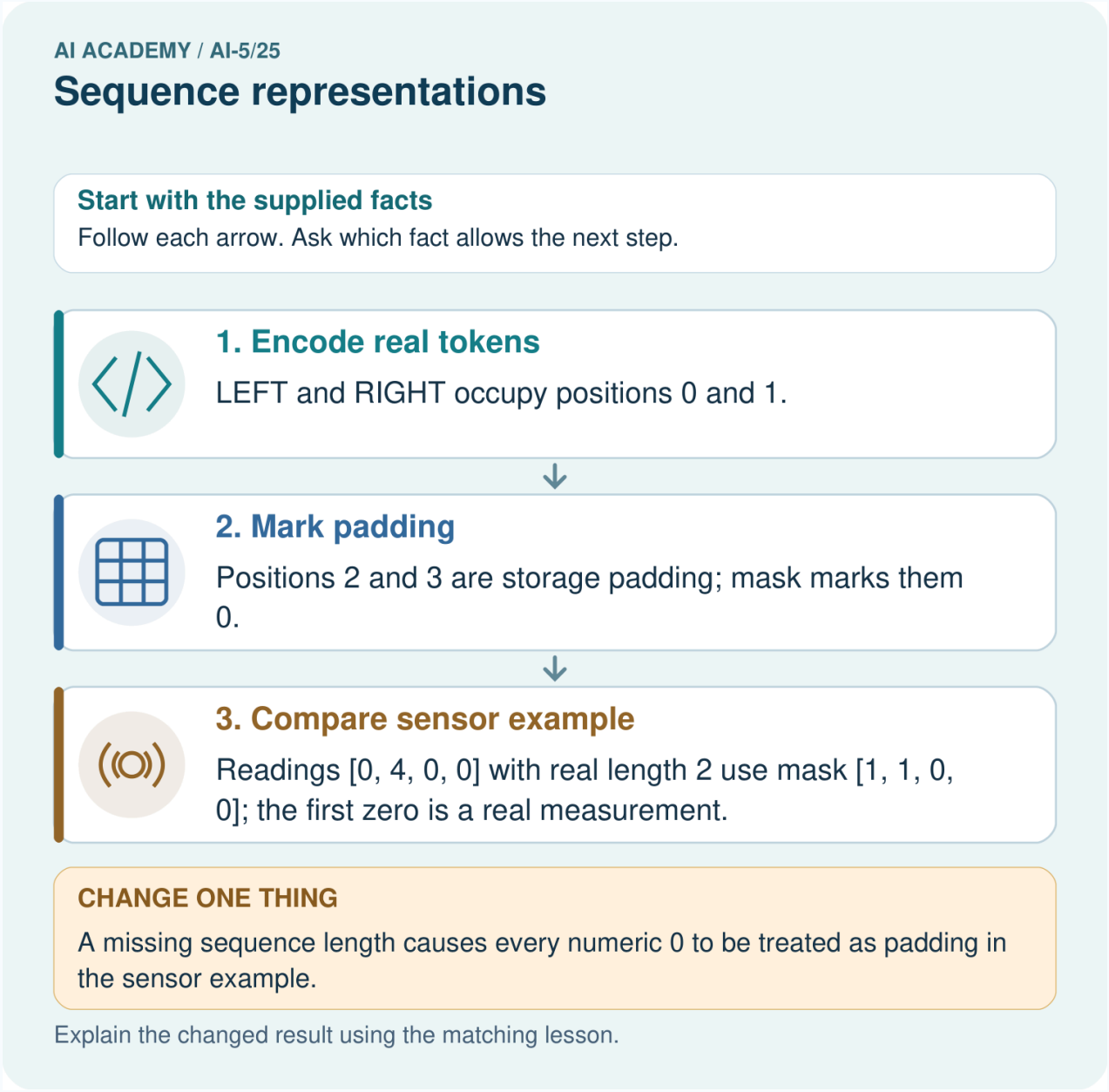
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Sequence representations

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

Teacher check: Teacher explanation: For example GO TURN STOP and STOP TURN GO both have count vector [1, 1, 1]. The first answer is Yes and the second No. Counts alone cannot distinguish them.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Sequence** representations. Proposed explanation: “Larger **token ID**s mean more of a concept”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: IDs identify symbols; their arithmetic sizes have no declared semantic meaning.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The explicit real length 2 or equivalent **mask**. Values alone cannot tell a genuine zero measurement from a **padding** zero. Changed-case check: The first real sensor reading is incorrectly discarded.

Its numeric value is 0, but its position is real according to the declared length.

Changed-case reasoning: The first real sensor reading is incorrectly discarded. Its numeric value is 0, but its position is real according to the declared length.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide a ledger with position, original symbol, ID and **mask**. Use separate coloured or labelled cards for real unknowns and placeholders without relying on colour alone.

Extension: enumerate all logs of length one through four over GO, TURN and STOP, verify encode-and-decode recovery, and find count-vector collisions with different final-event answers.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve representation collisions and compare GO STOP with STOP GO.
7 to 18 minutes	Declare the vocabulary and task, then complete W1 and W2.
18 to 32 minutes	Trace the encoder, including a real unknown event and its mask.
32 to 45 minutes	Use W3 to distinguish lost unknown identity from artificial padding.
Session 2 start to 8 minutes	Complete the guided final-position calculation and compare stored versus real length.
8 to 20 minutes	Calculate masked sensor means, including a measured zero, and explain the empty-mask boundary.
20 to 35 minutes	Collect W4 independently, checking the new vocabulary and five-position layout.
35 to 45 minutes	Use W5 and the exit response to audit truncation, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main encoding/order and sensor probes as supplied. W2GO TURN STOPversusSTOP TURN GObothcounts[1, 1, 1]withfinalSTOPversusGO. W3WAIT/PAUSE GOcollision loses identity, not real length. W5 rejectfiveevents/emptyundercontract; truncationchangesfinalevent. IndependentIDs[2, 4, 3, 0, 0], mask[1, 1, 1, 0, 0], lastrealindex 2 containsCLOSE. Sensor maskedmean(0+6+3)/3=3, all-slot 9/5=1.8.

Zero is a real measurement and stays in denominator. Extended code's review result for lastUNKis a declared policy, not proof of recovered event identity.

Diagnose the current misconceptions

Larger **token ID**s mean more of a concept: IDs identify symbols; their arithmetic sizes have no declared semantic meaning.

Zero always means missing data: A measured zero can be real; the **mask** determines which positions contribute.

A **mask** automatically fixes every model computation: The relevant operation must actually use the **mask**, and later aggregation can still discard order.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

Calculate their **mask**ed mean and the incorrect mean over all five slots.

Teacher explanation: IDs [2, 4, 3, 0, 0], **mask** [1, 1, 1, 0, 0]; final real position 2 contains CLOSE. **Mask**ed sensor mean is (0 + 6 + 3) / 3 = 3. Un**mask**ed mean is 9 / 5 = 1.8. The measured zero must count in the denominator.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Encode and recover the final event

Encode GO TURN STOP using the lesson vocabulary and four positions. Give its **mask**, real length and final real event.

Teacher explanation: IDs are [2, 3, 4, 0], **mask** [1, 1, 1, 0], real length 3 and final real event STOP at position 2.

W2 Expose an order collision

Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

Teacher explanation: For example GO TURN STOP and STOP TURN GO both have count vector [1, 1, 1]. The first answer is Yes and the second No. Counts alone cannot distinguish them.

W3 Separate unknown from padding

Encode WAIT GO. Which positions are real? Compare it with PAUSE GO, and state what information the encoding loses.

Teacher explanation: Both encode as [1, 2, 0, 0] with **mask** [1, 1, 0, 0]. Positions 0 and 1 are real. UNK at position 0 loses whether the symbol was WAIT or PAUSE; **padding** is only at positions 2 and 3.

W5 Audit length and empty input

The encoder receives five events ending in STOP, then an empty log. What should it do under the lesson contract? Why should a program not simply crop the first and assume a final event for the second?

Teacher explanation: It rejects both: one exceeds four real events and the other has none. Cropping can remove the final event and change the task answer.

An empty log has no final real position; the program needs an explicit policy rather than an invented event.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: TURN STOP becomes IDs [3, 4, 0, 0] and **mask** [1, 1, 0, 0]. Length is 2, so final real position is 1 with ID 4, STOP. Position 3 is PAD.

Exit: the full five-event log answers Yes, but its first four events end in TURN and answer No. Rejection makes the unsupported length visible and avoids silently changing the question.

W1 Encode and recover the final event

W2 Expose an order collision

W3 Separate unknown from padding

W4 Apply a fresh schema independently

Use PAD = 0, UNK = 1, OPEN = 2, CLOSE = 3, CHECK = 4 and five stored positions. Encode OPEN CHECK CLOSE and give its **mask** and final real position. Separately, sensor values [0, 6, 3, 0, 0] have that same **mask**.

Calculate their **mask**ed mean and the incorrect mean over all five slots.

Teacher explanation: IDs [2, 4, 3, 0, 0], **mask** [1, 1, 1, 0, 0]; final real position 2 contains CLOSE. **Mask**ed sensor mean is (0 + 6 + 3) / 3 = 3. Un**mask**ed mean is 9 / 5 = 1.8. The measured zero must count in the denominator.

W5 Audit length and empty input

Independent assessment and fresh retry

W4 requires IDs [2, 4, 3, 0, 0], **mask** [1, 1, 1, 0, 0], final index 2, **mask**ed mean 3 and un**mask**ed mean 1.8. Fresh retry: use the same five-position vocabulary for CHECK OPEN.

IDs are [4, 2, 0, 0, 0], **mask** [1, 1, 0, 0, 0], final index 1 is OPEN. Sensor values [8, 0, 0, 0, 0] with that **mask** have mean 4, not 8 or 1.6.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **sequence** representations **evidence** sheet to Image-pattern Network Investigation. Show how you can encode an ordered **sequence** with an explicit vocabulary.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide a ledger with position, original symbol, ID and **mask**. Use separate coloured or labelled cards for real unknowns and placeholders without relying on colour alone.

Extension: enumerate all logs of length one through four over GO, TURN and STOP, verify encode-and-decode recovery, and find count-vector collisions with different final-event answers.

Week 26 explores embeddings and neighbourhoods, including why a vector representation is different from an arbitrary **token ID**.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/26, ai-5/28, ai-5/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	

Record	Teacher evidence
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Encode an ordered **sequence** with an explicit vocabulary

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Give two logs made of exactly GO, TURN and STOP once each that have equal count vectors but different answers to whether the final event is STOP.

Reasoned practice answer: Teacher explanation: For example GO TURN STOP and STOP TURN GO both have count vector [1, 1, 1]. The first answer is Yes and the second No. Counts alone cannot distinguish them.

Check every learner before discussion: Use the worked case for **Sequence** representations. Proposed explanation: “Larger **token ID**s mean more of a concept”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: IDs identify symbols; their arithmetic sizes have no declared semantic meaning.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Require a **prediction** and a reason before comparing.

Expected result: The first real sensor reading is incorrectly discarded. Reason: Its numeric value is 0, but its position is real according to the declared length.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

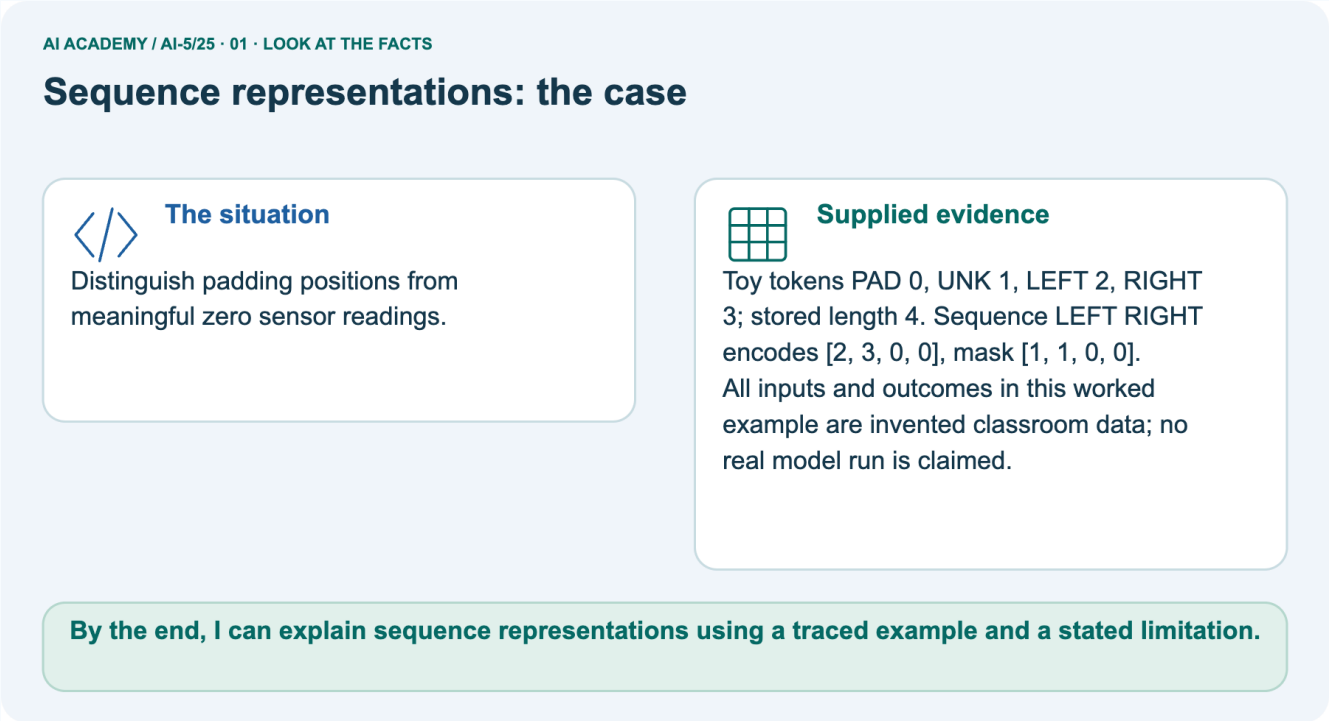
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Sequence representations: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

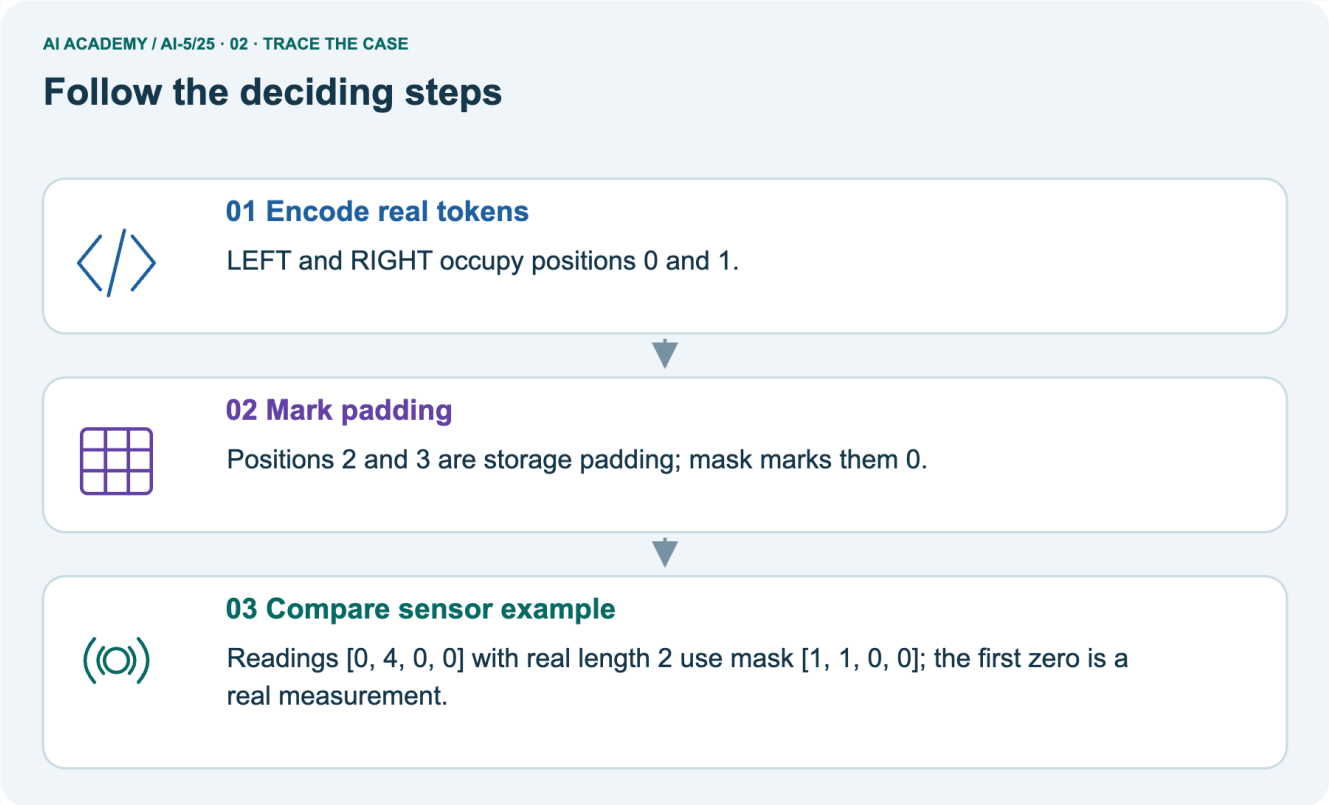
Distinguish **padding** positions from meaningful zero sensor readings. Supplied facts: Toy tokens PAD 0, UNK 1, LEFT 2, RIGHT 3; stored length 4. **Sequence** LEFT RIGHT encodes [2, 3, 0, 0], **mask** [1, 1, 0, 0].

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Encode real tokens: LEFT and RIGHT occupy positions 0 and 1. → Mark **padding**: Positions 2 and 3 are storage **padding**; **mask** marks them 0. → Compare sensor example: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. A padded log still has two real events



AI ACADEMY / AI-5/25 · 03 · INSPECT THE LESSON MODEL

A padded log still has two real events

Stored position	Token ID	Mask
0	2 meaning GO	1 real
1	4 meaning STOP	1 real
2	0 meaning PAD	0 placeholder
3	0 meaning PAD	0 placeholder

The final stored slot is padding. For this right-padded contiguous mask, the real length is the mask sum, and the final real event is at length minus one. A different masking layout would need a different indexing rule.

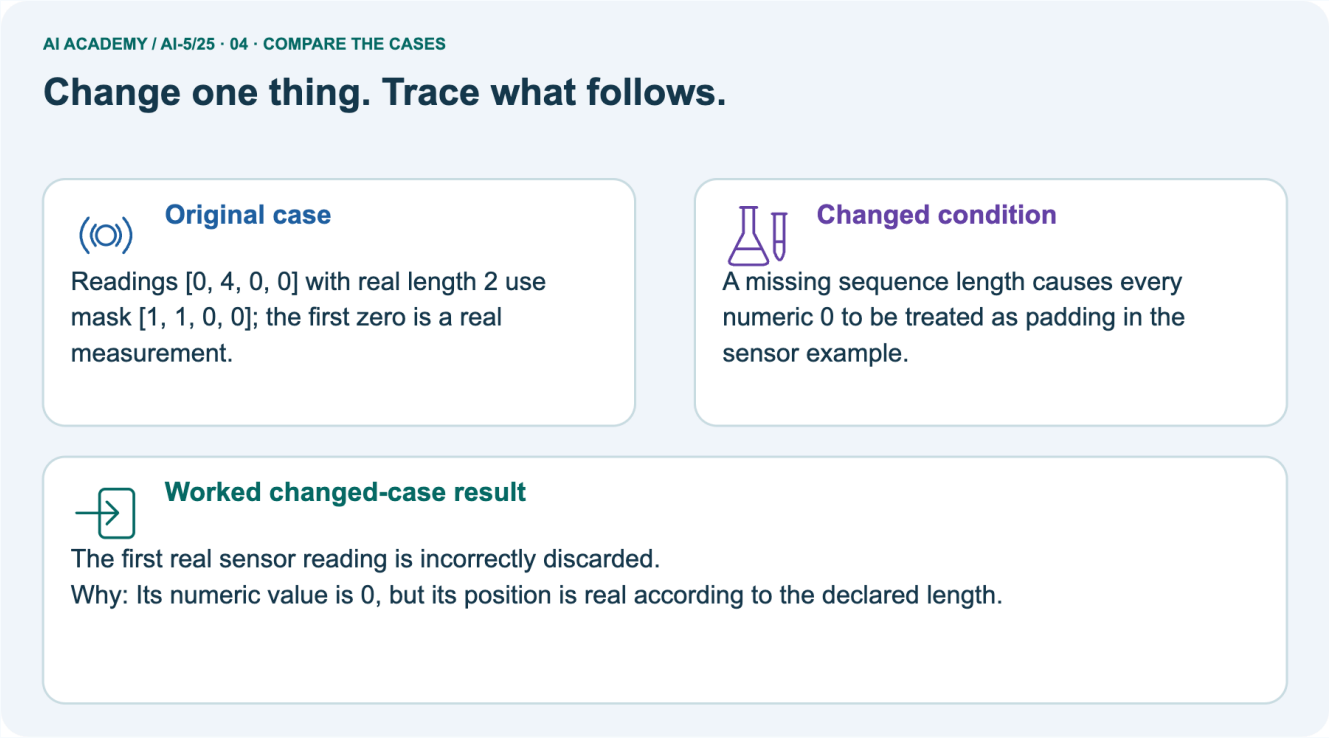
Follow the labelled picture with the matching explanation. Open the vector diagram

A padded log still has two real events. 0 | 2 meaning GO | 1 real. 1 | 4 meaning STOP | 1 real. 2 | 0 meaning PAD | 0 placeholder. 3 | 0 meaning PAD | 0 placeholder

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Readings [0, 4, 0, 0] with real length 2 use **mask** [1, 1, 0, 0]; the first zero is a real measurement.

Changed condition: A missing **sequence** length causes every numeric 0 to be treated as **padding** in the sensor example. Expected result: The first real sensor reading is incorrectly discarded.

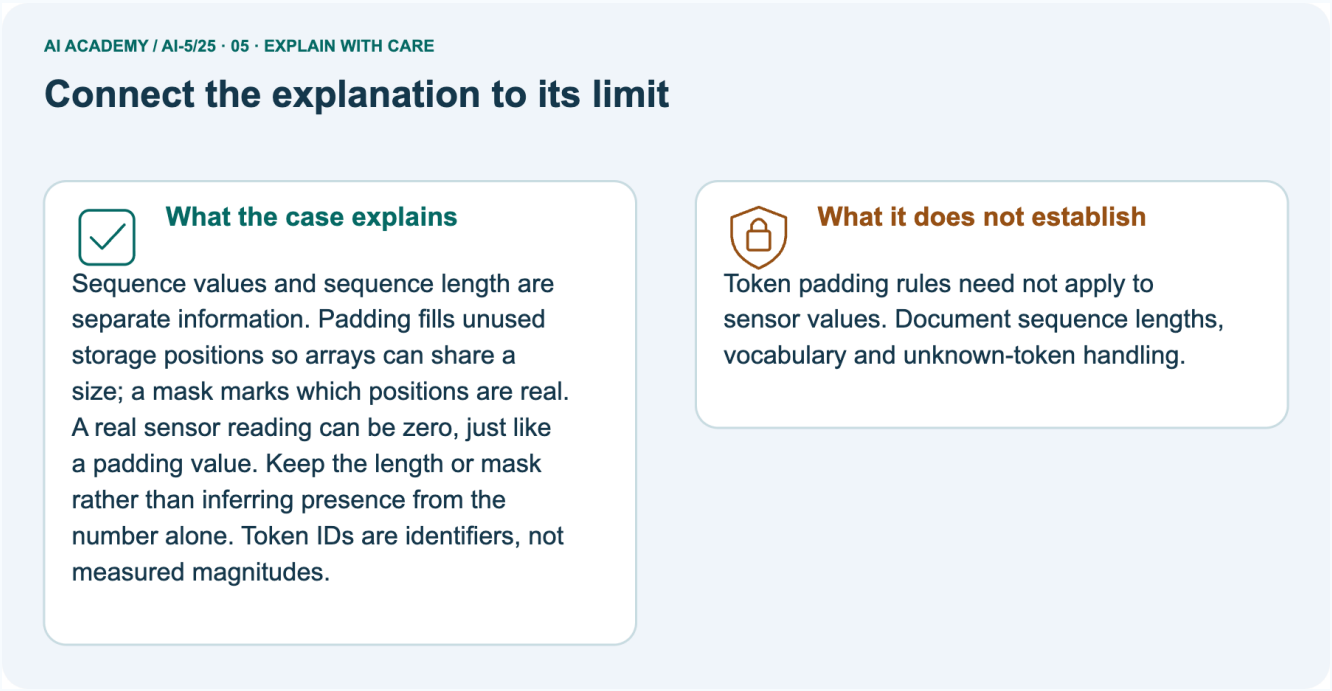
Reason: Its numeric value is 0, but its position is real according to the declared length.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Its numeric value is 0, but its position is real according to the declared length.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Sequence values and **sequence** length are separate information. **Padding** fills unused storage positions so arrays can share a size; a **mask** marks which positions are real.

A real sensor reading can be zero, just like a **padding** value. Keep the length or **mask** rather than inferring presence from the number alone. **Token ID**s are identifiers, not measured magnitudes.

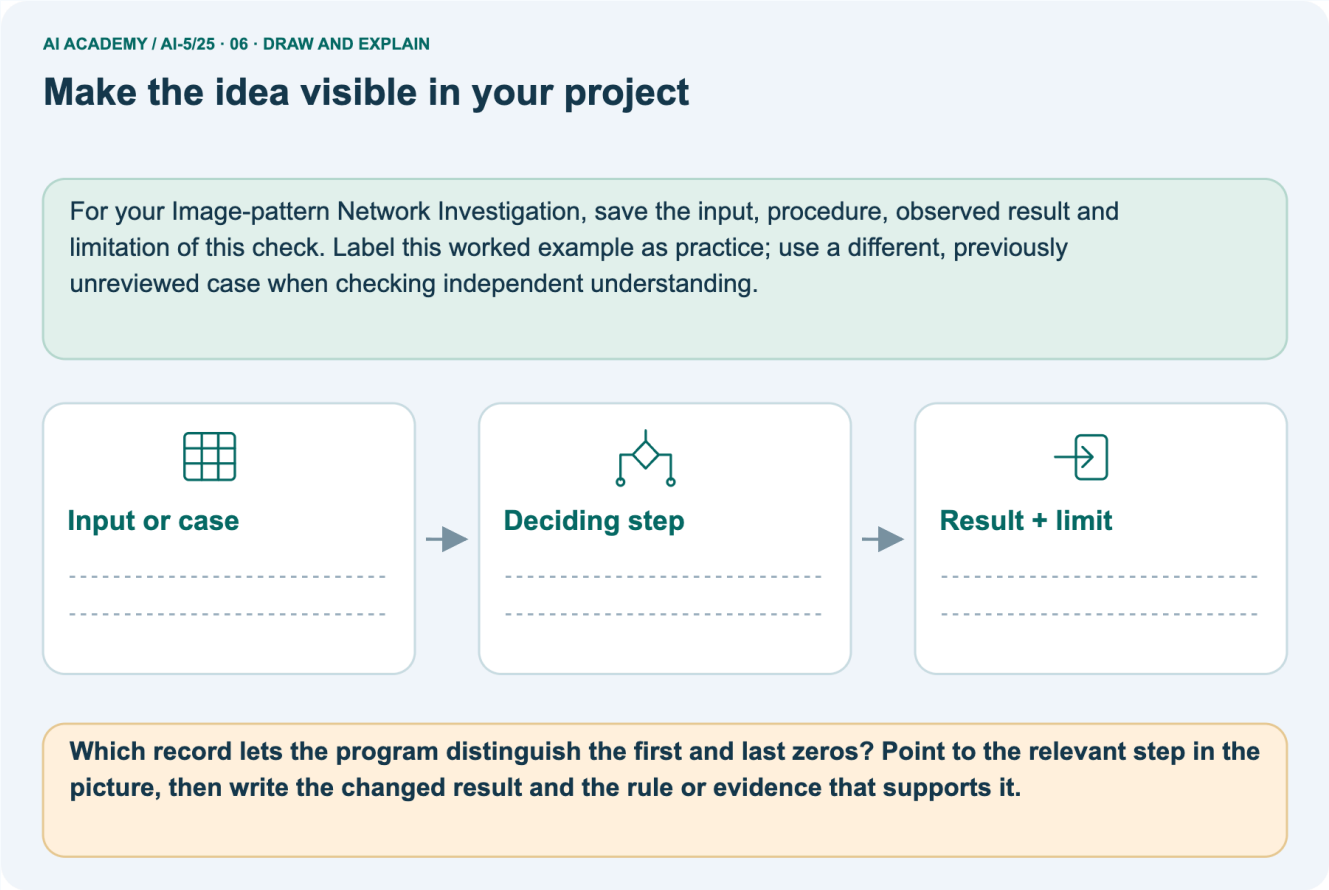
Limit: Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Token **padding** rules need not apply to sensor values. Document **sequence** lengths, vocabulary and unknown-token handling.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Which record lets the program distinguish the first and last zeros? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The explicit real length 2 or equivalent **mask**. Values alone cannot tell a genuine zero measurement from a **padding** zero. Changed-case check: The first real sensor reading is incorrectly discarded.

Its numeric value is 0, but its position is real according to the declared length.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Sequence representations: the case

02. Follow the deciding steps

03. A padded log still has two real events

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.